Box 62.1 Components of the Dialysis Prescription

Duration
Frequency
Vascular access
Dialyzer (membrane, configuration, surface area, sterilization method)
Blood flow rate
Dialysate flow rate
Ultrafiltration rate
Dialysate composition (see Table 62.7)
Anticoagulation
Dialysate temperature
Intradialytic medications

Box_62_1

Box 62.2 Factors Causing Malnutrition in Hemodialysis Patients

Inadequate protein or calorie intake
Increased energy expenditure
Metabolic acidosis
Hormonal alterations
Comorbidities or hospitalizations
Inflammation
Dialytic nutrient losses
Dialysis-induced catabolism
Infection

Box_62_2

Box 62.2a Global Nutritional Assessment Tools to Evaluate for Presence of Protein Energy Wasting (PEW)—Subjective Global Assessment (SGA), Malnutrition Inflammation Score (MIS), and PEW Score

	SGA	*MIS*	*PEW*
Medical History			
Weight change	X	X	
Dietary intake	X	X	
Gastrointestinal symptoms	X	X	
Functional capacity	X	X	
Metabolic requirements/Comorbidity	X	X	
Physical Examination			
Signs of fat wasting	X	X	
Signs of muscle wasting	X	X	
Edema due to poor nutrition	X		
BMI (kg/m^2)		X	X
Laboratory			
Serum albumin (g/L)		X	X
Serum TIBC (mg/dL)		X	
Serum Cr (mg/dL normalized for BSA)			X
nPNA (g/kg/day)			X

Box_62_2a

Box 62.3 Interventions to Consider in Recurrent Intradialytic Hypotension

Reassess dry weight
Reduce interdialytic sodium gain
Reduce intradialytic sodium gain
Assess intradialytic hypocalcemia, hypokalemia, and hypomagnesemia
Avoid food intake during dialysis
Adjust antihypertensive medications and their timing
Assess cardiac function
Cool dialysate
Extend dialysis time or add sessions per week
Use sequential ultrafiltration or ultrafiltration modeling
Prescribe midodrine

Box_62_3

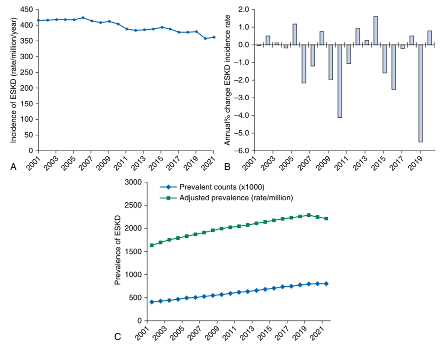


Fig. 62.1 (A) Adjusted incidence rates of end-stage kidney disease *(ESKD)* in the United States with time. (B) Annual percentage change in adjusted incidence rates of ESKD with time. (C) Prevalent counts (×1000) and adjusted prevalence of ESKD in the United States with time. (Modified from U.S. Renal Data System. 2023 USRDS Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2023.)

Fig_62_1

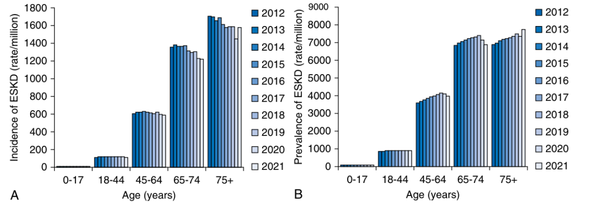


Fig. 62.2 (A) Incidence and (B) prevalence of end-stage kidney disease *(ESKD)* (rate/million by year) with age from 2012 to 2021. (Modified from U.S. Renal Data System. 2023 USRDS Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2023.)

Fig_62_2

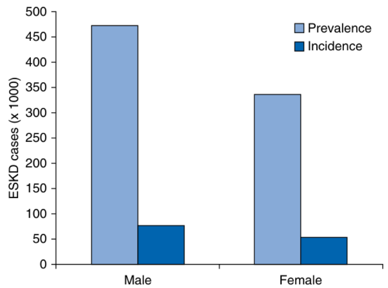


Fig. 62.3 Incidence and prevalence of end-stage kidney *(ESKD)* disease by sex in 2020. (Modified from U.S. Renal Data System, 2022 Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 20892.)

Fig_62_3

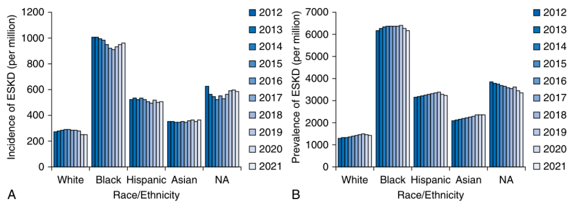


Fig. 62.4 (A) Incidence and (B) prevalence of end-stage kidney disease *(ESKD)* with race and ethnicity (rate/million by year). *Black,* Black or African American; *NA,* Native American. (Modified from U.S. Renal Data System. 2023 USRDS Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2023.)

Fig_62_4

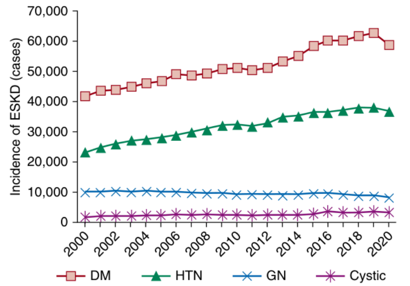


Fig. 62.5 Causes of end-stage kidney disease (ESKD) in the United States. (Modified from U.S. Renal Data System. 2022 USRDS Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 20892.)

Fig_62_5

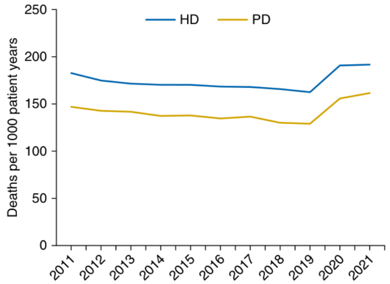
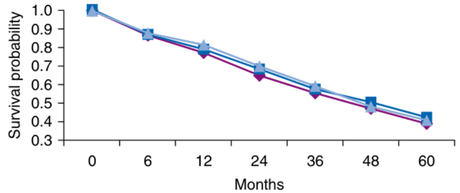


Fig. 62.6 Mortality rates for patients on dialysis. (Modified from U.S. Renal Data System. 2023 USRDS Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2023.)

Fig_62_6



	0	6	12	24	36	48	60
2007	1	0.86	0.77	0.65	0.55	0.47	0.39
2012	1	0.87	0.79	0.68	0.58	0.50	0.42
2017	1	0.88	0.81	0.70	0.59	0.48	0.40

Fig. 62.7 Probability of survival for hemodialysis patients by cohort year. (Modified from U.S. Renal Data System. 2023 USRDS Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2023.)

Fig_62_7

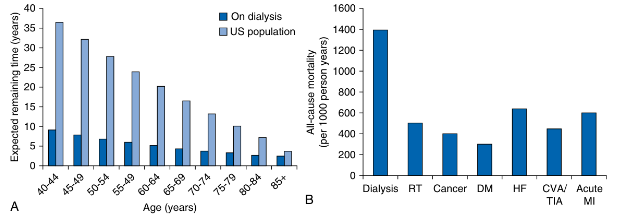


Fig. 62.8 (A) Life expectancy and (B) adjusted all-cause mortality rates per 1000 person years for end-stage kidney disease (ESKD) patients in 2021. *CVA*, Cerebrovascular accident; *DM*, diabetes mellitus; *HF*, heart failure; *MI*, myocardial infarction; *RT*, renal transplant; *TIA*, transient ischemic attack. (Modified from U.S. Renal Data System. 2023 USRDS Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2023.)

Fig_62_8

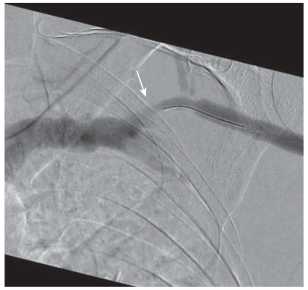


Fig. 62.9 Angiography of a cephalic arch stenosis *(white arrow)* that developed in a patient with a left upper arm AV fistula. The high blood flow rate in a location where the vein turns 90 degrees leads to turbulent flow, leading to intimal hyperplasia. Lesions like this need repeated angioplasty and sometimes benefit from stent placement or use of drug-eluting balloons. (Photo courtesy of Dr. Suresh Appasamy, Interventional Nephrology, University of California, Davis.)

Fig_62_9

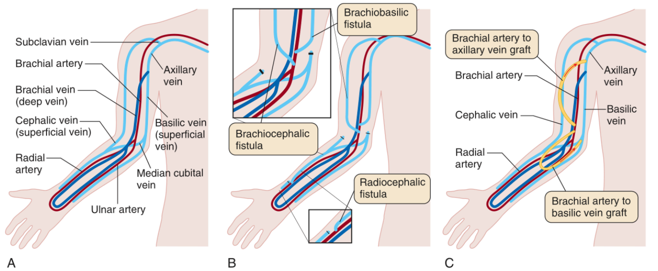


Fig. 62.10 (A) Vasculature of the right upper limb outlining superficial and deep veins and major arteries. (B) Location of anatamoses for typical wrist (radiocephalic) and upper arm (brachiocephalic and brachiobasilic) arteriovenous (AV) fistulae. (C) Location of typical AV grafts.

Fig_62_10

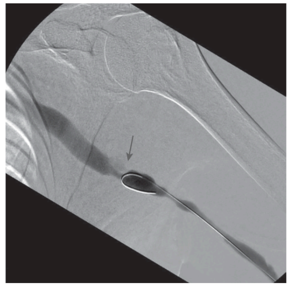


Fig. 62.11 Angiography of stenosis in the outflow vein of a left upper arm AV graft *(gray arrow).* This typical lesion in patients with an AV graft is felt due to turbulent flow as blood moves from graft to vein. These areas of intimal hyperplasia generally respond to angioplasty but tend to recur, oftentimes requiring stenting or drug-eluting balloons to maintain long-term patency. There are also numerous intragraft stenoses. (Photo courtesy Dr. Suresh Appasamy, Interventional Nephrology, University of California, Davis.)

Fig_62_11

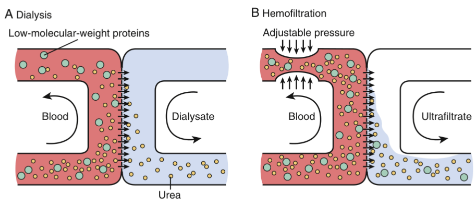


Fig. 62.12 Diffusion versus convection. (A) Diffusion across a semipermeable membrane. The driving force for solute diffusion is the transmembrane concentration gradient. Small solutes with higher concentrations in the blood compartment, such as potassium, urea, and small uremic toxins, diffuse through the membrane into the dialysate compartment. Dialysis dissipates this concentration gradient (i.e., the molecular concentration gradient decreases with dialysis). Larger solutes and low-molecular-weight proteins, such as albumin, diffuse poorly across the semipermeable membrane. (B) Convection across a semipermeable membrane. The driving force for convection, commonly called ultrafiltration, is the transmembrane hydrostatic pressure. When applied to the blood compartment, solvent flows across the membrane into the dialysate compartment, bringing along solutes. For solutes with a sieving coefficient close to 1, there is no change in concentrations in the blood compartment with time. (From Meyer TW, Hostetter TH. Uremia. *N Engl J Med.* 2007;357(13):1316–1325.)

Fig_62_12

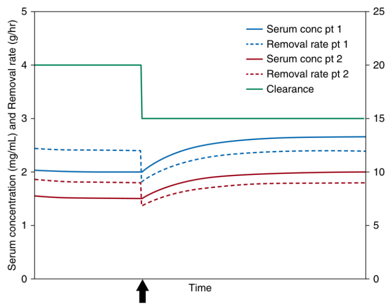


Fig. 62.13 Clearance versus removal rate of a substance is illustrated here, in two patients with the same 36 L volume of distribution of urea with the same constant clearance of 20 mL/min. Patients 1 and 2 have urea generation rates of 40 g/hour and 30 g/hour, respectively. When the clearance of both patients suddenly drops to 10 mL/min, absolute removal rate of urea drops but at a different rate despite the same change in clearance. See text for more information.

Fig_62_13

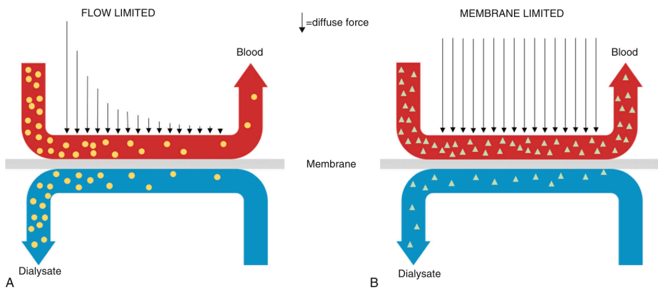


Fig. 62.14 (A) Flow-limited clearance. A logarithmic decline in small solute concentrations, indicated by the *arrows* on either side of the membrane, is depicted from the dialyzer inlet at the *left* to the blood outlet on the *right*. This decline is based on rapid diffusion of solute across the membrane and forms the basis for Eq. 5. Removal is maximized by countercurrent flow of blood and dialysate. (B) Membrane-limited clearance: The diffusive force is still the solute concentration gradient. Solute concentrations along the membrane from dialyzer inflow to outflow for both blood and dialysate are relatively constant because transport across the membrane is limiting and relatively low.

Fig_62_14

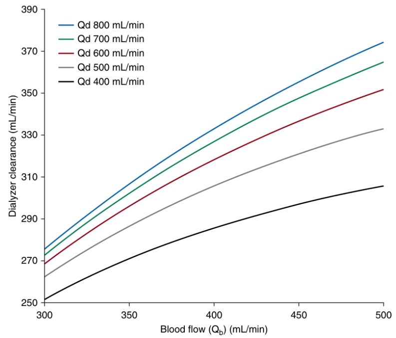


Fig. 62.15 For small, flow-limited solute removal, like urea, across a membrane, an increase in blood flow rate increases dialyzer clearance to a much greater extent than a similar increase in dialysate flow rate. This model assumes no convective clearance from ultrafiltration and assumes a dialyzer mass transfer-area coefficient (*K_0A*) for urea of 1000, like what we expect for a modern, high-flux dialyzer.

Fig_62_15

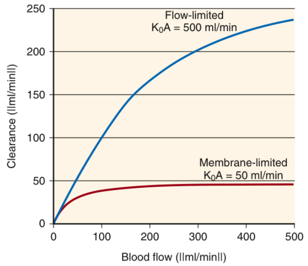


Fig. 62.16 Flow-limited versus membrane-limited clearances. Flow limits clearance when the membrane is not fully exposed to inflow solute concentrations (see Fig. 62.12). This typically occurs for small, easily dialyzed solutes. In contrast, larger solutes tend to saturate the membrane along its entire length at lower blood (or dialysate) flow rates (see Fig. 62.13). For these less easily dialyzed solutes, further increases in flow have no influence on clearance. *K_0A*, Mass transfer-area coefficient.

Fig_62_16

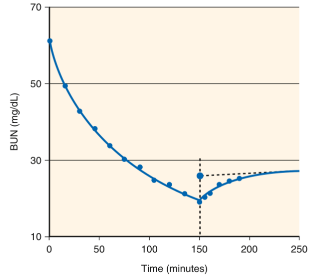


Fig. 62.17 Equilibrated postdialysis blood urea nitrogen *(BUN)*, the basis for eKt/V. Precise measurements of BUN every 15 minutes during a typical patient's 2.5-hour hemodialysis shows a logarithmic decrease in concentration during the treatment and a rapid rebound that is complete approximately 1 hour later. The double-pool model shown in Fig. 62.23 and the *solid line* in the graph predict the concentrations accurately. The equilibrated postdialysis BUN is an extrapolated value shown as the *large solid circle*.

Fig_62_17

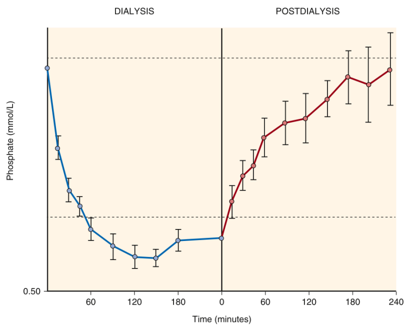


Fig. 62.18 Effect of sequestration on serum phosphate levels and removal during dialysis. Measurements of serum inorganic phosphorus concentrations were taken at 15-minute intervals during both high-flux and standard hemodialysis. The rapid flux of phosphorus due to its relatively high mass transfer-area coefficient caused levels to fall into the hypophosphatemic range (below the lower *dotted line*) during most of the 4-hour dialysis. A marked rebound continued for 4 hours after dialysis ended. (Modified from DeSoi CA, Umans JG. Phosphate kinetics during high-flux hemodialysis. *J Am Soc Nephrol.* 1993;4:1214–1218.)

Fig_62_18

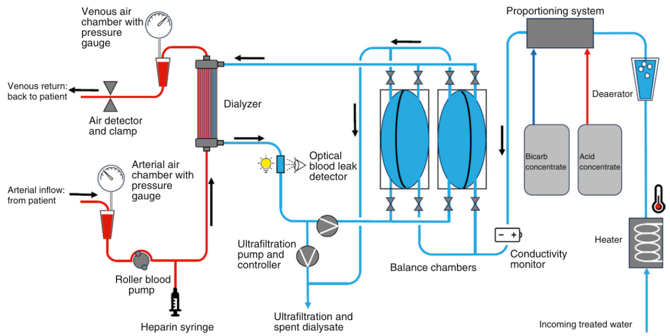


Fig. 62.19 Components of a typical modern hemodialysis blood and dialysate circuit. Much of the dialysate circuit is not readily seen but is internal to the machine. A typical "3 stream" dialysate (water, acid, and bicarbonate concentrates) proportioning system and the use of precise balance chambers to ensure accurate volumetric control of ultrafiltration highlight the advances in modern dialysis machinery.

Fig_62_19

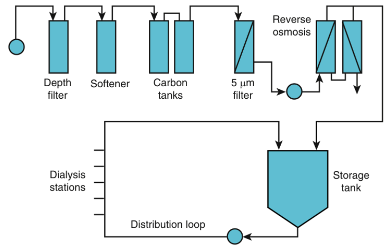


Fig. 62.20 Schematic of a typical configuration of a reverse osmosis water treatment system. Tap water undergoes filtration to remove gross particulate matter and then is softened before exposure to charcoal (carbon tanks) to remove contaminants such as chloramine. A second filtration process removes particulate matter, as well as microbiologic organisms. Finally, water is filtered under high pressure to remove dissolved contaminants such as aluminum (reverse osmosis). Product water is then either stored in a water tank or piped directly to each dialysis station.

Fig_62_20

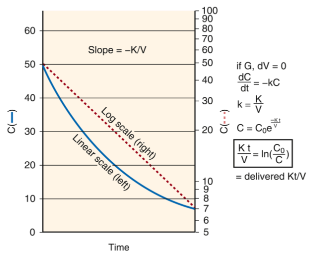


Fig. 62.21 Intradialysis urea kinetics: origin of Kt/V. *Left,* The nonlinear decrease in blood urea nitrogen during dialysis *(solid line)* becomes a straight line when plotted on a log scale *(dashed line).* The fractional rate of decrease is a constant, k = K/V, where k is the elimination constant, K is the clearance, and V is the urea distribution volume. *Right,* The solution to the equation describing first-order kinetics shows that delivered Kt/V primarily depends on the predialysis and postdialysis blood urea nitrogen values (see text and legend for Fig. 62.20 for definition of variables shown). This oversimplified equation is expanded in Fig. 62.20 to include the other important variables.

Fig_62_21

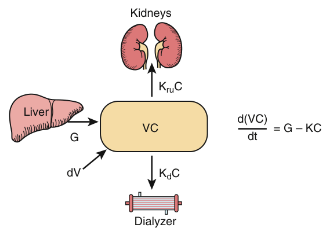


Fig. 62.22 Single-pool model of urea mass balance in a hemodialysis (HD) patient. When the patient is not undergoing dialysis (most of the time for conventional HD), K_d is 0 and removal is determined solely by K_{ru}. V is the urea distribution volume, equated to body water space; C is urea concentration; and dV is the rate of fluid gain (negative during dialysis, positive between dialyses). During HD, total clearance (K) is the sum of K_d and K_{ru}. An explicit solution is available to the differential equation that describes the rate of urea accumulation or loss (dVC/dt) as the difference between generation *(G)* and removal *(KC).*

Fig_62_22

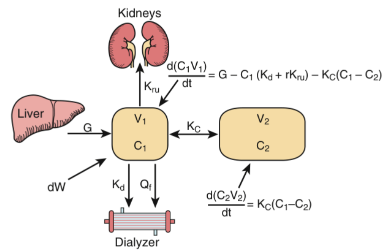


Fig. 62.23 Double-pool model of urea mass balance in a hemodialysis patient. Addition of a second compartment to the diffusion model of urea mass balance shown in Fig. 62.22 accounts for the postdialysis rebound in urea concentration shown in Fig. 62.17 and, in general, is considered a more accurate model. K_c is the coefficient of mass transfer between compartments, analogous to dialyzer K_0A. Solution of the differential equation requires numeric analysis and is not commonly applied in dialysis clinics.

Fig_62_23

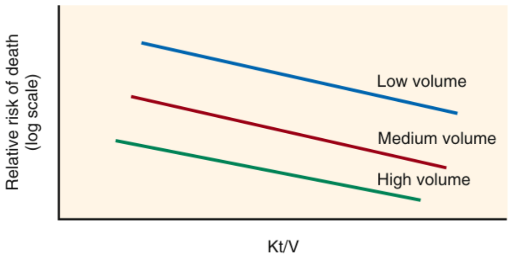


Fig. 62.24 Risk of death as a function of dialysis dose and body size. The risk of death in hemodialysis patients decreases with increased dialysis dose *(Kt/V)* and may be further stratified by urea volume as a measure of body size. Larger patients, in general, have a lower death risk.

Fig_62_24

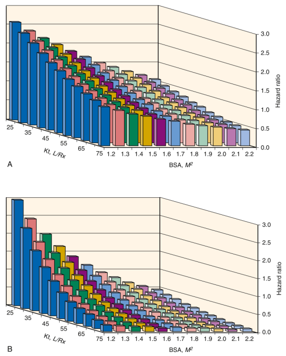


Fig. 62.25 Risk of mortality related to body size and dialysis dose. These data were obtained from a large observational study of 43,334 patients. (A) Hazard ratio analysis was adjusted for case mix. (B) Hazard ratio analysis included an interaction term between Kt and body mass index and was adjusted for case mix. *BSA,* Body surface area; *L/Rx,* liters per treatment. (From Lowrie EG, Li Z, Ofsthun N, Lazarus JM. Body size, dialysis dose and death risk relationships among hemodialysis patients. *Kidney Int.* 2002;62:1891–1897.)

Fig_62_25

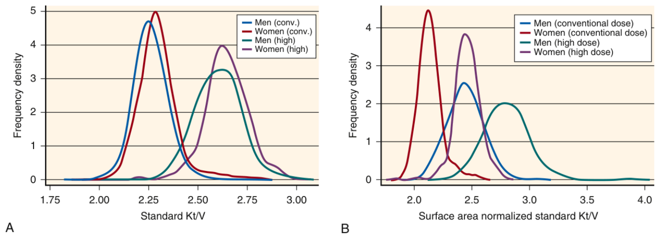


Fig. 62.26 (A) Standard Kt/V in the conventional and high-dose HEMO study subjects, by sex. (B) Surface area normalized standard Kt/V in the conventional and high-dose HEMO study subjects, by sex. Conversion to surface area was based on an anthropometric estimate of V in each patient.[320,739] *V,* Urea distribution volume. (From Daugirdas JT, Greene T, Chertow GM, Depner TA. Can rescaling dose of dialysis to body surface area in the HEMO study explain the different responses to dose in women versus men? *Clin J Am Soc Nephrol.* 2010;5(9):1628–1636.)

Fig_62_26

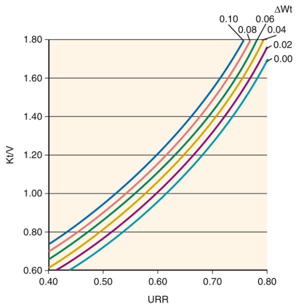


Fig. 62.27 Curvilinear relationship between urea reduction ratio *(URR)* and Kt/V, stratified by degree of ultrafiltration during dialysis. Whereas the URR (see text) falls with increasing fluid removal during dialysis (ΔWt) from 0% to 10% of body weight, Kt/V increases. The latter more appropriately accounts for the increase in clearance caused by ultrafiltration. Curves are derived from formal urea modeling.

Fig_62_27

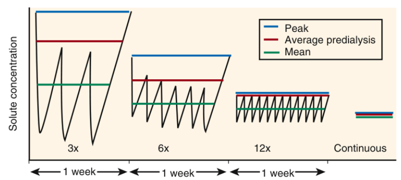


Fig. 62.28 Effect of frequency on peak and average solute concentrations. Two-compartment formal kinetic modeling of a solute with low K_c predicts that both peak and mean concentrations decrease significantly as the frequency of treatments increases, despite no change in the weekly dialyzer clearance × treatment time (Kt).

Fig_62_28

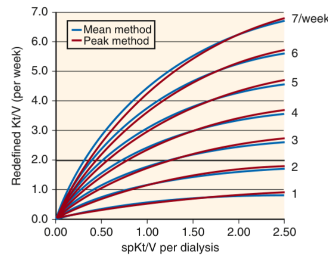


Fig. 62.29 Effect of increased dose versus increased frequency on effective clearance. The effective clearance, expressed as "standard Kt/V" on the vertical axis, tends to plateau, despite increases in dialyzer clearance, expressed as single-pool Kt/V (spKt/V) on the horizontal axis. Two different models of solute kinetics show similar diminishing returns as the delivered dose increases. A marked increase in effective clearance can be achieved only by increasing the frequency of treatments.

Fig_62_29

Table 62.1 Causes of End-Stage Kidney Disease in Incident and Prevalent Patients[a]

	Incident Patients		Prevalent Patients	
Primary Kidney Disease	**No.**	**% of Total**	**No.**	**% of Total**
Diabetes mellitus	59,474	45.6	309,030	38.3
Hypertension	37,168	28.5	213,824	26.5
Glomerulonephritis	8395	6.4	117,603	14.6
Cystic kidney disease	3397	2.6	41,042	5.1
Other urologic	1679	1.3	17,997	2.2
Other/Unknown cause	20,409	15.6	108,424	13.4
Total	130,522	100	807,920	100

[a]In the United States in 2020.
Data from U.S. Renal Data System, 2022 Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 20892.

Table_62_1

Table 62.2 Causes of Death for Prevalent Hemodialysis Patients (2021)

Cause of Death	% of Total
Cardiovascular disease	38%
• Arrhythmia, cardiac arrest	32%
• Myocardial infarction	2%
• Other cardiovascular	4%
Withdrawal from dialysis	12%
COVID-19	7%
Septicemia or other infection	6%
Malignancy	2%
Other causes	8%
Missing or unknown causes	27%

Data from U.S. Renal Data System. 2023 USRDS Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2023.

Table_62_2

Table 62.3 Characteristics of Ideal Hemodialysis Vascular Access Compared With Commonly Available Types

Desired Characteristic	Autogenous AV Fistula	AV Graft	Central Venous Catheter
Good primary patency rate	✓	✓✓	✓✓✓
Instant usability	✓	✓✓	✓✓✓
Long survival	✓✓✓	✓✓	✓
Low thrombosis rate	✓✓✓	✓✓	✓
Low infection rate	✓✓✓	✓✓	✓
High blood flow rate on hemodialysis	✓✓✓	✓✓✓	✓
Patient comfort	✓	✓	✓✓✓
Patient bathing/hygiene	✓✓✓	✓✓✓	✓
Minimizes needles	✓	✓	✓✓✓
Minimal cosmetic affect	✓	✓✓	✓✓

AV, Arteriovenous.

Table_62_3

Table 62.4 Factors Influencing Effective Clearance

	Treatment-Related (in Order of Importance)	
Solute-Related	**Small Molecules**	**Large Molecules**
Molecular size	Blood and dialysate flow	Membrane permeability
Molecular charge	Membrane surface area	Treatment time
Macromolecular binding	Treatment time	Membrane surface area
Body distribution and sequestration	Membrane permeability	Blood and dialysate flow

Table_62_4

Table 62.5 Key Factors That Affect the Solute Clearance of a Hemodialyzer

Parameter	Key Factors	Clearance
Properties of the membrane	Membrane porosity	↑
	Membrane thickness	↓
	Membrane surface area	↑
	Membrane charge	Varies
	Membrane hydrophilicity	↑
Properties of the solute	Increased molecular weight and size	↓
	Charge	Varies
	Lipid solubility	↓
	Protein binding	↓
Blood side	Unstirred blood layer	↓
	Increased blood flow	↑
Dialysate side	Dialysate channeling and unstirred layer	↓
	Increased dialysate flow	↑
	Countercurrent direction of flow	↑

Table_62_5

Table 62.6 Characteristic Values for Standard, High-Efficiency, and High-Flux Dialyzers

	Standard	High Efficiency	High Flux
Blood flow rate (mL/min)	250	≥350	≥350
Dialysate flow rate (mL/min)	500	≥500	≥500
K_0A urea	300–500	≥600	Variable
Urea clearance (mL/min)	<200	>210	Variable
Urea clearance/body weight (mL/min/kg)	< 3	>3	Variable
Vitamin B_{12} clearance (mL/min)	30–60	Variable	>100
β-2 microglobulin clearance (mL/min)	<10	Variable	>20
Ultrafiltration coefficient or K_{UF} (mL/h/mm Hg)	3.5-5.0	Variable	>20
Membrane material	Cellulose	Variable	Variable

Table_62_6

Table 62.7 Solutes Present in the Dialysate

Solute	Concentration (mEq/L)
Sodium	135-145
Potassium	0-4
Chloride	102-106
Bicarbonate	25-40
Acetate	2-8
Calcium	0-3.5
Magnesium	0.5-1.0
Dextrose	100-200 (mg/dL)

Table_62_7

Table 62.8 Guidelines for Anticoagulation in Hemodialysis Patients at High Risk for Serious Bleeding

Anticoagulation for Hemodialysis	Clinical Condition
No anticoagulation or regional anticoagulation	Active bleeding
	Recent intracerebral hemorrhage
	Significant coagulopathy or thrombocytopenia
	Major surgery within 7 days
	Intracranial surgery within 14 days
	Biopsy of visceral organ within 72 hours
	Pericarditis
Low-dose heparin	Major surgery beyond 7 days
	Biopsy of visceral organ beyond 72 hours
	Minor surgery 8 hours prior
	Minor surgery within 72 hours
Low-dose heparin or no anticoagulation	Major surgery 8 hours prior

Table_62_8